Quality Assessment Scoring for Primary Studies

This document presents the detailed results of the Quality Assessment (QA) phase conducted for the **98** studies at the full text review stage from which **82** were included in this Systematic Literature Review. Each study was evaluated against five technical criteria (QA1–QA5) designed to assess problem clarity, algorithmic rigor, data fidelity, evaluation transparency, and critical discussion. To ensure objectivity and methodological consistency, scoring was performed independently by both researchers using a standardized 0–1 scale, with a minimum cumulative threshold of **3.5 out of 5.0** required for final inclusion. The following table maps the quality performance of each selected study.

QA1. How clearly do the authors define and present the insider threat detection problem in the ML context?
QA2. How effectively do the authors describe the ML models, algorithms, and learning paradigms used for insider threat detection?
QA3. How well does the study describe the source of the dataset, its characteristics, and its distribution (e.g., class balance)?
QA4. How do the authors report the performance of ML models using standard metrics (e.g., accuracy, precision, recall, F1-score, and Area Under the Curve AUC)?
QA5. How well do the authors discuss the limitations, challenges (e.g. XAI or adversarial training), and future work of their ML-based models??

	Author (Year)	QA1	QA2	QA3	QA4	QA5	Total Score	Decision
1	Adun (2023)	1.0	1.0	0.5	0.5	0.5	3.5	Include
2	Alabdulkareem (2022)	1.0	1.0	1.0	1.0	0.5	4.5	Include
3	Ahmadi (2025)	1.0	0.5	0.5	1.0	1.0	4.0	Include
4	Alhammadi 2021	1.0	1.0	0.5	1.0	1.0	4.5	Include
5	Ali 2025	1.0	5.0	1.0	1.0	1.0	4.5	Include
6	Ahmed 2025	0.5	1.0	1.0	1.0	0.5	4.0	Include
7	ALmihqani et al (2021)	1.0	1.0	0.5	1.0	0.0	3.5	Include
8	almusawi (2024)	1.0	1.0	1.0	1.0	0.5	4.5	Include
9	AL-Mihqani (2022)	1.0	1.0	0.5	1.0	1.0	4.5	Include
10	Al-Shehari et al. (2024)	1.0	1.0	1.0	0.0	1.0	4.0	Include
11	Al-Shehari et al. (2023)	1.0	1.0	1.0	1.0	1.0	5.0	Include
12	Al-Shehari et al. (2024)	1.0	1.0	1.0	1.0	1.0	5.0	Include
	AlQadheeb et al. (2022)	05	0.0	1.0	0.0	0.5	2.0	Exclude
13	Al-Shehari et al. (2023)	1.0	1.0	1.0	1.0	0.5	4.5	Include
14	Alshehri Abdul (2022) Relational DL	1.0	1.0	1.0	1.0	1.0	5.0	Include
15	Amiri-Zarandi (2023)	1.0	1.0	1.0	1.0	1.0	5.0	Include
16	Al-Shehari (2021)	1.0	1.0	1.0	1.0	1.0	5.0	Include
17	Amuda (2022)	1.0	1.0	1.0	1.0	0.5	4.5	Include
18	Anju (2024)	1.0	1.0	1.0	1.0	1.0	5.0	Include
19	Anakath (2022)	1.0	1.0	1.0	1.0	1.0	5.0	Include
20	Asha S (2023)	1.0	1.0	1.0	1.0	1.0	5.0	Include
21	Cai X (2024)	1.0	1.0	1.0	1.0	1.0	5.0	Include
	Dahiya and Kumar (2024)	0.0	0.0	0.0	0.5	1.0	1.5	Exclude
22	Dong et al. (2025)	1.0	1.0	1.0	1.0	1.0	5.0	Include
23	Eshmawi (2026)	1.0	1.0	1.0	1.0	1.0	5.0	Include
24	Feng (2025)	1.0	1.0	1.0	1.0	1.0	5.0	Include
25	Ferraro (2025)	1.0	1.0	1.0	1.0	1.0	5.0	Include
26	Gayathri, R.G. (2025) advers	1.0	1.0	1.0	1.0	1.0	5.0	Include
27	Gayathri, B. (2025) Cloud	1.0	1.0	0.0	1.0	1.0	4.0	Include
28	Gayathri, R.G. (2024) SPCGAN	1.0	1.0	1.0	1.0	1.0	5.0	Include
29	Gonzales (2025)	1.0	1.0	0.0	1.0	1.0	4.0	Include
30	Gupta (2024)	1.0	1.0	1.0	1.0	1.0	5.0	Include
31	Hafizu Rhman (2022)	1.0	1.0	1.0	1.0	1.0	5.0	Include
32	Haq et al (2022)	1.0	1.0	1.0	0.5	1.0	4.5	Include
33	He Daojing et al (2022)	1.0	1.0	1.0	1.0	1.0	5.0	Include
34	He Daojing et al (2024)	1.0	1.0	1.0	1.0	1.0	5.0	Include
35	Huang (2025)	1.0	1.0	0.5	0.5	0.5	3.5	Include
36	Jaiswal (2024)	1.0	1.0	1.0	1.0	0.5	4.5	Include
37	Janjua (2021)	1.0	1.0	1.0	0.5	0.5	4.0	Include
38	Kamatchi et al. (2025)	1.0	0.5	1.0	0.5	0.5	3.5	Include
39	Kong et al. (2025)	1.0	1.0	1.0	1.0	0.5	4.5	Include
40	Kotb et al. (2025)	1.0	1.0	1.0	1.0	0.5	4.5	Include
41	Lavanya et al. (2024)	1.0	1.0	1.0	1.0	0.5	4.5	Include
42	Lavanya et al. (2025)	1.0	1.0	1.0	1.0	0.5	4.5	Include
43	Le & Zincir-Heywood (2021)	1.0	1.0	1.0	1.0	0.5	4.5	Include
44	Li et al. (2023) Graph Convolutional Network	1.0	1.0	1.0	1.0	1.0	5.0	Include
45	Li et al. (2024) Graph Meta-Learning	1.0	1.0	1.0	1.0	1.0	5.0	Include
	Li et al. (2024) Nuclear Security Reasoning	0.0	0.5	0.0	1.0	1.0	2.5	Exclude
46	Liu et al. (2025)	1.0	1.0	1.0	1.0	0.5	4.5	Include
47	Medvedev et al. (2025)	1.0	1.0	1.0	0.5	0.5	4.0	Include
48	Mehmood et al. (2023)	1.0	1.0	1.0	0.5	0.0	3.5	Include
49	Mehnaz & Bertino (2021)	1.0	1.0	1.0	0.5	0.0	3.5	Include

50	Mladenovic et al. (2024)	1.0	1.0	1.0	1.0	1.0	5.0	Include
51	Nasir et al. (2021)	1.0	1.0	1.0	1.0	0.5	4.5	Include
	Nasser et al. (2025) Mobile Crowd Sourcing	0.5	0.0	0.0	0.0	1.0	1.5	Exclude
52	Nikiforova et al. (2024)	1.0	0.5	1.0	0.0	1.0	3.5	Include
53	Pal et al. (2023)	1.0	1.0	1.0	1.0	1.0	5.0	Include
54	Patel & Iyer (2025)	1.0	1.0	0.5	1.0	0.0	3.5	Include
55	Peccatiello et al. (2023)	1.0	0.5	1.0	0.5	0.5	3.5	Include
56	Pennada et al. (2024)	1.0	0.5	1.0	0.5	0.5	3.5	Include
57	Pennada et al. (2025)	1.0	1.0	1.0	1.0	1.0	5.0	Include
58	Álvarez Muñiz et al. (2025)	1.0	1.0	1.0	1.0	1.0	5.0	Include
59	Qawasmeh & AlQahtani (2025)	1.0	1.0	1.0	1.0	1.0	5.0	Include
	Qiu et al. (2025)	0.0	0.5	0.5	0.0	0.5	1.5	Exclude
60	Randive et al. (2023)	1.0	1.0	1.0	1.0	1.0	5.0	Include
61	Rauf et al. (2021)	1.0	1.0	0.5	1.0	1.0	4.5	Include
	Rizvi & Williams (2024)	1.0	0.0	0.0	0.0	1.0	2.0	Exclude
62	Roy & Chen (2024)	1.0	1.0	0.5	1.0	1.0	4.5	Include
63	Sarhan & Altwaijry (2022)	1.0	0.5	1.0	1.0	1.0	4.5	Include
64	Senevirathna et al. (2025)	1.0	1.0	0.5	1.0	0.0	3.5	Include
65	Song et al. (2024)	1.0	1.0	1.0	1.0	0.0	4.0	Include
66	Tabassum et al. (2024)	1.0	0.5	1.0	1.0	0.0	3.5	Include
67	Tian T et al. (2025)	1.0	1.0	1.0	1.0	1.0	5.0	Include
68	Tian Z et al. (2024)	1.0	1.0	1.0	1.0	1.0	5.0	Include
69	Villarreal-Vasquez et al. (2023)	1.0	1.0	1.0	1.0	1.0	5.0	Include
70	Wall & Agrafiotis (2021)	1.0	1.0	1.0	1.0	1.0	5.0	Include
71	Wang & El Saddik (2023)	1.0	1.0	1.0	1.0	1.0	5.0	Include
72	Wang Zhi et al. (2024) FedITD	1.0	1.0	1.0	1.0	1.0	5.0	Include
73	Wang Jiarong et al. (2023) Deep Cluster	1.0	1.0	1.0	1.0	1.0	5.0	Include
74	Wei Yichen et al. (2021)	1.0	1.0	1.0	1.0	1.0	5.0	Include
75	Wei Zhiyuan et al. (2024)	1.0	1.0	1.0	1.0	1.0	5.0	Include
76	Wen et al. (2023)	1.0	0.5	1.0	1.0	1.0	4.5	Include
77	Xiao Junchao et al. (2023)	1.0	1.0	1.0	1.0	1.0	5.0	Include
78	Xiao Fengrui et al. (2025)	1.0	1.0	1.0	1.0	1.0	5.0	Include
79	Xiao Haitao et al. (2024)	1.0	1.0	1.0	1.0	1.0	5.0	Include
80	Ye Xiaoyun et al. (2025)	1.0	1.0	1.0	1.0	1.0	5.0	Include
81	Yildirim & Anarim (2022)	1.0	1.0	1.0	1.0	1.0	5.0	Include
82	Zhu et al. (2024)	1.0	1.0	0.5	1.0	1.0	4.5	Include

The list below has the 10 studies that were identified as secondary (survey or review) work:

	Author (Year)	Decision
1	Gong et al. (2024)	Exclude but keep as a reference
2	Kamatchi & Uma (2025)	
3	Alketbi & Mehmood (2025)	
4	Kundiya & Haribhakta (2025)	
5	Rauf et al. (2023)	
6	Yuan & Wu (2021)	
7	Alzaabi & Mehmood (2024)	
8	Aloraini et al. (2022)	
9	Al-Muntaser et al. (2023)	
10	Ofori et al. (2025)	